

Auditable Failure Memory for ROS 2 Navigation: An Exploratory Comparison with Spatial Caching

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Abstract

Autonomous mobile robots navigating dynamic indoor environments frequently encounter execution failures at physical chokepoints, narrow passages, and transient blockages. Standard navigation architectures (e.g., ROS 2 Nav2) employ reactive intra-episode recovery behaviors, yet they lack structured cross-episode memory indexing, leading to repetitive dead-end retries across subsequent tasks. While prior embodied agent frameworks (such as REFLECT) employ multi-modal experience summarization with LLMs for failure explanation, this work investigates event-driven failure recording, perception-driven invalidation, and offline replay auditing directly within the ROS 2 navigation stack. Specifically, we present **FailMem**, an event-driven, causally bound failure memory architecture for autonomous mobile robots. FailMem records immutable observation bundles comprising raw LiDAR range arrays, TF coordinate transforms, and raw 2D costmap regions-of-interest (ROIs), binding navigation failures to specific execution regions and action identities while dynamically invalidating suppression upon verified sensory clearance. To ensure rigorous scientific evaluation and eliminate unverified synthetic abstractions, we implement an independent replay audit pipeline that recomputes sensory metrics directly from raw physical observations against SHA256 checksum manifests. Evaluated across 30 physical simulation runs in Gazebo 11 with ROS 2 Humble Nav2 ($n = 3$ per condition under sequential fixed-seed execution), FailMem (F) eliminates redundant dead-end traversals relative to a constrained reactive baseline (R), reducing total travel distance by -18.5% (14.03 m vs. 17.22 m) and total execution time by -22.3% (109.5 s vs. 141.0 s). Furthermore, dynamic memory invalidation saves -11.0% total distance relative to permanent suppression ($M1$) upon environmental recovery (16.25 m vs. 18.25 m). In our exploratory single-agent benchmark under static 2D geometric blockages, FailMem and a spatial observation caching baseline (O) exhibited identical high-level route choices with $< 1\%$ differences in reported total travel distance and total execution duration (14.03 m vs. 13.95 m in D1; 16.25 m vs. 16.40 m in D2). In this exploratory dataset, route choices were identical and showed no additional benefit of F over O ; general scenarios and statistical equivalence have not been verified. Finally, an exploratory feasibility check ($n = 4$) on candidate action-conditioned execution (H_1) revealed that local planners negotiated narrow inflation margins without controller abortion (4/4 Nav2 **SUCCEEDED**, 3/4 verified strict physical arrival), indicating that the tested scenario did not establish executability divergence and providing concrete guidance for future action-profiled memory designs.

1. Introduction

Autonomous mobile robots deployed in complex indoor environments—such as fulfillment warehouses, clinical facilities, and office buildings—must repeatedly traverse structured corridors and shared doorways. In such environments, unmapped physical blockages (e.g., temporarily parked carts or closed security doors) frequently obstruct the nominal shortest path. Standard navigation systems, such as the ROS 2 Navigation Stack (Nav2) (Steve Macenski et al. 2020; Steven Macenski et al. 2022), rely on layered 2D costmaps (Lu, Hershberger, and Smart 2014) and local recovery behaviors (e.g., in-place rotations, backup maneuvers, costmap clearing). While these recovery routines assist in negotiating transient obstacles during active trajectory tracking, they are bound to the execution lifetime of individual navigation actions. When a new navigation goal is dispatched from a distant decision junction, the robot has no structured representation of prior execution failures in occluded corridors, resulting in unguided retry traversals into known dead ends before local sensors can re-acquire line-of-sight.

To mitigate unguided re-exploration, two primary paradigms have been explored:

1. **Spatial Occupancy Mapping:** Updating global geometric representations (e.g., Costmap2D (Lu, Hershberger, and Smart 2014), OctoMap (Hornung et al. 2013), Voxelblox (Oleynikova et al. 2017)) as sensor observations reveal occupied space.
2. **Episodic Failure Memory & Verbal Reflection:** Storing execution traces, error diagnostics, or natural language self-reflections (e.g., Reflexion (Shinn et al. 2023), REFLECT (Liu, Bahety, and Song 2023)) to inform subsequent planning.

However, existing frameworks present key operational and methodological limitations. In embodied AI, verbal reflection mechanisms typically store unstructured natural language strings in prompt contexts without physical sensor postcondition verification or spatial costmap expiration rules. Furthermore, robotics evaluation historically suffers from reproducibility challenges when relying on synthetic 2D mock state machines that omit real controller oscillation, sensor noise, and lifecycle timing delays.

In this work, we present an auditable, event-driven failure memory framework and evaluate its behavior in physical ROS 2 / Gazebo simulation. Specifically, we provide:

1. **Authentic Action Client & Perception Architecture:** We implement FailMem on genuine ROS 2 Humble Nav2 action clients, capturing immutable observation bundles (untruncated laser scans, TF transforms, and raw 2D costmap ROI subgrids) prior to state classification.
2. **Independent Replay Audit Pipeline:** We implement an independent offline replay engine that reconstructs memory lifecycles and physical trajectories directly from raw serialized sensor records against SHA256 checksum manifests, validating chronological causality ($t_{\text{rec}} \geq \max(t_{\text{act}}, t_{\text{obs}}) - 0.05 \text{ s}$) and physical halt stability.
3. **Empirical Evaluation Across 30 Physical Runs:** Across 30 physical simulation runs in Gazebo 11 ($n = 3$ per condition), we demonstrate that historical knowledge eliminates redundant dead-end exploration (-18.5% distance vs. Reactive R) and dynamic invalidation prevents permanent detour traps (-11.0% distance vs. Persistent $M1$).
4. **Calibrated Comparison with Spatial Caching:** We report that under static 2D geometric blockages, FailMem and spatial caching (O) produce identical route choices with $< 1\%$ differences in reported total travel distance and total execution duration. In this exploratory dataset, route choices were identical and showed no additional benefit of F over O ; general scenarios and statistical equivalence have not been verified. An exploratory feasibility trial ($n = 4$) on action-conditioned navigation (H_1) further shows that the candidate scenario did not establish executability divergence, clarifying the empirical boundaries for future action-profiled memory designs ($F2$).

2. Related Work & Mechanism Taxonomy

2.1 Spatial Occupancy Mapping & Dynamic Distance Fields

Layered 2D costmaps (Lu, Hershberger, and Smart 2014; Steve Macenski et al. 2020) maintain geometric obstacles by ray-casting sensor ranges into grid cells and propagating configuration-space inflation. 3D volumetric frameworks, such as OctoMap (Hornung et al. 2013), employ hierarchical octrees with discrete probabilistic voxel occupancy updates. In contrast, distance-field frameworks such as Voxelblox (Oleynikova et al. 2017) incrementally construct continuous Euclidean Signed Distance Fields (ESDF) and Truncated Signed Distance Fields (TSDF) for smooth trajectory optimization. While spatial mapping accurately maintains geometric free and occupied space, it treats unobserved areas identically and does not capture action-specific execution failures (e.g., narrow doorpost inflation boundaries where space is free but specific controller trajectories abort).

2.2 Execution Monitoring & Plan Diagnosis

Deliberative robotics frameworks (Fikes and Nilsson 1971; Simmons et al. 1998; Ingrand and Ghallab 2017) formalize plan execution monitoring and diagnostic repair. Modern navigation stacks (Steve Macenski et al. 2020) execute recovery behaviors when local planners fail. However, standard recovery states are discarded upon goal termination, requiring explicit cross-phase or cross-task episodic memory stores to guide high-level routing.

2.3 LLM Self-Reflection & Agent Memory

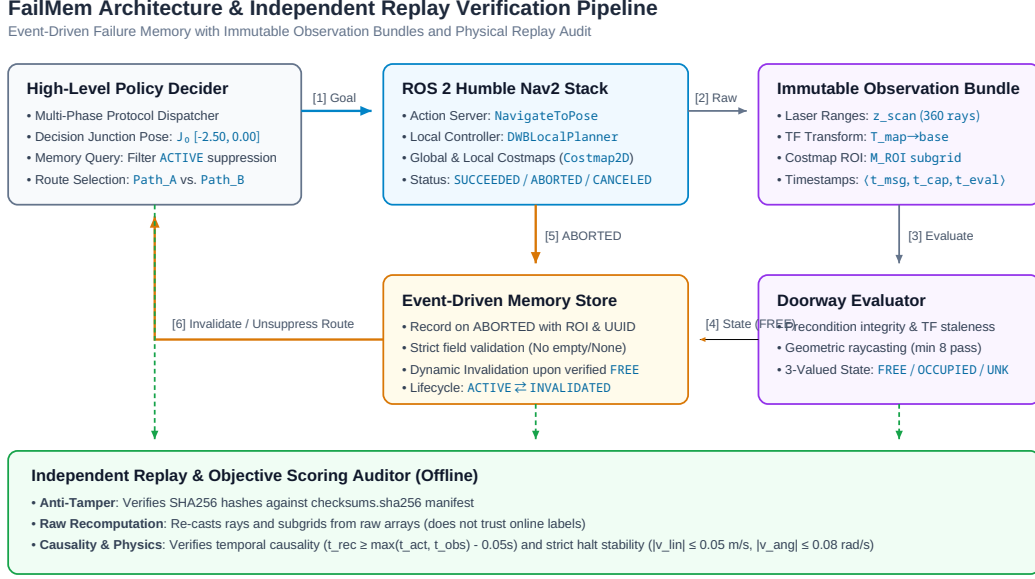
Recent embodied AI systems investigate large language models for planning and error correction. Reflexion (Shinn et al. 2023) prompts LLMs with scalar task feedback to generate natural language self-reflections. Inner Monologue (Huang et al. 2023) integrates success detectors and scene descriptions into closed-loop prompt chains. REFLECT (Liu, Bahety, and Song 2023) uses hierarchical multi-modal experience summarization with LLMs to generate structured failure explanations and plan corrections. While these methods demonstrate prompt-level flexibility, they focus on higher-level task reflection. FailMem complements these approaches by investigating event-driven failure recording, perception-driven invalidation, and offline replay auditing directly within the low-level ROS 2 action and costmap stack.

2.4 Experience-Based Planning & No-Good Learning

In motion planning, the Lightning framework (Berenson, Srinivasa, and Kuffner 2012) and Experience Graphs (E-Graphs) (Phillips et al. 2012) reuse libraries of prior successful paths to accelerate search. In discrete optimization, conflict-driven clause learning (e.g., GRASP (Marques-Silva and Sakallah 1999)) records “no-goods” from search dead ends to prune infeasible branches. FailMem adapts the principle of negative constraint recording to mobile robot navigation with explicit sensor-verified invalidation.

Comparative Mechanism			
Taxonomy			
Paradigm	Representation	Grounding Level	Invalidation
Trigger			
Costmap2D / Nav2 traversal	2D Occupancy Grid	Geometric cells	Direct laser ray
OctoMap traversal	3D Voxel Octree	Discrete 3D voxels	Direct laser ray
Voxblox integration	Continuous ESDF / TSDF	3D Distance Field	Dynamic voxel
Experience Graphs repair	Path Waypoints / Graph	Discrete Path	Collision check on
Reflexion reset	Natural Language Text	Semantic tokens	Manual context
REFLECT termination	Natural Language Text	Multi-modal logs	Session
FailMem (Implemented) FREE state	Event Tuple	Raw sensor bundle	Sensor-verified
FailMem F2 (Proposed) check	Profile Tuple	Bundle + Profile	Profile capability

3. The FailMem Architecture & Verification Pipeline



FailMem Architecture and Verification Pipeline

3.1 Immutable Observation Bundles

Prior to evaluating passage clearance, FailMem constructs an immutable observation bundle B :

$$B = \langle \text{obs_id}, \mathbf{z}_{\text{scan}}, \mathbf{T}_{\text{map} \rightarrow \text{base}}, \mathbf{M}_{\text{costmap_ROI}}, t_{\text{msg}}, t_{\text{capture}}, t_{\text{eval}} \rangle$$

where $\mathbf{z}_{\text{scan}}$ is the raw range array, $\mathbf{T}_{\text{map} \rightarrow \text{base}}$ is the TF transformation, and $\mathbf{M}_{\text{costmap_ROI}}$ is the extracted 2D subgrid matrix.

3.2 Geometric & Costmap Clearance Formulation

The passage state is evaluated according to the following strict hierarchical rules implemented in `src/doorway_evaluator.py`:

1. Precondition Integrity Checks:

- TF translation and yaw must be valid finite numbers.
- Timestamp alignment: $|t_{\text{tf}} - t_{\text{scan}}| \leq 0.80 \text{ s}$.
- Data freshness: $t_{\text{sim}} - t_{\text{tf}} \leq 1.0 \text{ s}$ and $t_{\text{sim}} - t_{\text{scan}} \leq 1.5 \text{ s}$.
- Ray projection: Number of valid finite rays $N_{\text{valid}} > 0$.
- Visibility: Number of rays intersecting doorway bounding box $N_{\text{intersect}} > 0$.
- Costmap status (if provided): $N_{\text{total_cells}} > 0$. If any precondition fails, $\text{State}(B) = \text{UNKNOWN}$.

2. Classification Rules:

$$\text{State}(B) = \begin{cases} \text{OCCUPIED}, & \text{if } N_{\text{hits_inside}} \geq 5 \text{ or } N_{\text{costmap_occupied_cells}} > 0 \\ \text{FREE}, & \text{if } N_{\text{hits_inside}} = 0 \text{ and } N_{\text{pass_through}} \geq 8 \text{ and } \text{CostmapCleared}(B) \\ \text{UNKNOWN}, & \text{otherwise} \end{cases}$$

where $\text{CostmapCleared}(B) \iff (N_{\text{unknown_cells}} = 0 \wedge N_{\text{occupied_cells}} = 0)$ within the doorway ROI $[-0.15, 0.15, 0.95, 1.45]$.

3.3 Memory Lifecycle & Invalidation Mechanics

- **Failure Registration:** When chokepoint traversal fails in an occupied region, an active entry is committed:

$$M = \langle \text{mem_id}, t_{\text{fail}}, \text{goal_uuid}, \text{action_id}, \text{region_id}, \mathbf{M}_{\text{costmap_ROI}} \rangle$$

- **Dynamic Invalidation:** When subsequent perception produces $\text{State}(B) = \text{FREE}$ with evidence timestamp $t_{\text{free}} > t_{\text{fail}}$, the memory entry is marked **INVALIDATED**, unsuppressing the route.
- **Cross-Phase Scope Clarification:** In the current experimental implementation, history collection, memory creation, decision dispatch, and memory reset occur within a single experimental runner session across sequential phases. This establishes phase-to-phase memory transfer within an experimental run, and has not yet been benchmarked as a persistent multi-task database across separate OS daemon lifetimes.

3.4 Independent Replay Audit Pipeline

To eliminate reliance on self-reported runtime summaries, the independent replayer (`scripts/replay_and_score_p2c.py`): 1. Verifies SHA256 hashes of all serialized files against `checksums.sha256`; 2. Re-computes subgrid ROI cell counts and 3-valued perception directly from raw `subgrid_matrix` and laser scans; 3. Integrates continuous odometry trajectory samples to verify path length and physical halt stability; 4. Enforces chronological causality with a 0.05 s tolerance window:

$$t_{\text{rec}} \geq \max(t_{\text{tr1_end}}, t_{\text{obs_eval}}) - 0.05 \text{ s}, \quad t_{\text{inv}} \geq \max(t_{\text{fail}}, t_{\text{free_eval}}) - 0.05 \text{ s}$$

Replay rejects empty strings, missing fields, and "unknown" placeholder values.

4. Experimental Setup & Benchmarking Protocol

4.1 Physical Simulation Environment & Geometry

Simulations are conducted in ROS 2 Humble with Gazebo 11 simulating a TurtleBot3 Waffle robot in an asymmetric dual-path environment (`configs/p2c_dualpath_world.model`). - **Path A (North Short Path):** Protocol waypoint polyline length is 6.124 m from Decision Junction $J_0[-2.50, 0.00]$ to Goal $[2.50, 0.00]$ through a 0.80 m doorway chokepoint at $y = 1.20$. - **Path B (South Bypass Detour):** Protocol waypoint polyline length is 9.105 m from J_0 to Goal via open corridor at $y = -2.40$.

4.2 Distance Metric Definitions

To prevent ambiguity across different evaluation granularities, distance metrics are defined as follows: 1. **Protocol Waypoint Polyline Length:** Ideal piecewise linear distance connecting defined route waypoints (Path A: 6.124 m, Path B: 9.105 m). 2. **Decision-Phase Traversed Distance:** Actual physical odometry distance traversed during Phase 2 after goal dispatch from Decision Junction J_0 (Path B direct: ≈ 7.54 m; Path A direct: ≈ 5.89 m). 3. **Total Episode Distance:** Cumulative odometry distance spanning both the Phase 1 historical observation/attempt leg and the Phase 2 decision navigation leg (14.03 m for D1 F/O, 17.22 m for D1 R, 16.25 m for D2 F, 18.25 m for D2 M1).

4.3 Evaluated Scenarios & Methods

- **Scenario D0 (Unobstructed Baseline):** Doorway is clear; robot navigates Path A without prior history.
- **Scenario D1 (Confirmed Blockage):** Doorway is blocked by a physical box obstacle. Phase 1 traverses from J_0 to chokepoint approach waypoint $[-1.50, 1.20]$, encounters blockage, and retreats to J_0 . Phase 2 evaluates routing dispatched from J_0 .

- **Scenario D2 (Cleared / Restored):** Following D1, the obstacle is removed. Phase 1 observes FREE doorway clearance from vantage waypoint $[-1.50, 1.20]$ and retreats to J_0 . Phase 2 evaluates routing dispatched from J_0 .

Evaluated Methods: 1. **Reactive Baseline (R):** No cross-phase memory. In D1, enters Path A approach leg to gate waypoint $[-1.50, 1.20]$, detects blockage, and executes structured fallback retreat to J_0 before taking Path B. All methods share identical Nav2 costmap configurations. 2. **Spatial Observation Cache (O):** Caches 2D geometric occupancy observed from vantage waypoint $[-1.50, 1.20]$ during Phase 1. Evaluated at J_0 during Phase 2 route dispatch. 3. **FailMem (F):** Event-driven failure memory with dynamic perception-driven invalidation. 4. **Persistent Suppression ($M1$):** Static failure memory permanently suppressing Path A without dynamic invalidation.

4.4 Execution Constraints & Halt Stability

- **Sample Size & Order:** $n = 3$ per condition, executed consecutively under fixed simulation seeds.
- **Budget:** Total episode simulation budget is 180.0 s.
- **Physical Halt Stability:** Upon goal completion, the robot must maintain linear velocity $|v_{lin}| \leq 0.05 \text{ m/s}$ and angular velocity $|v_{ang}| \leq 0.08 \text{ rad/s}$ over a nominal 2.0 s stability recording window (with 2.3 – 2.4 s sim time coverage in the active executor).

5. Empirical Results & Boundary Analysis

5.1 Condition-Level Aggregated Metrics

Table 1 presents condition-level performance across 30 physical simulation runs ($n = 3$ per condition, reporting mean $\pm$ sample standard deviation with Bessel correction $ddof = 1$).

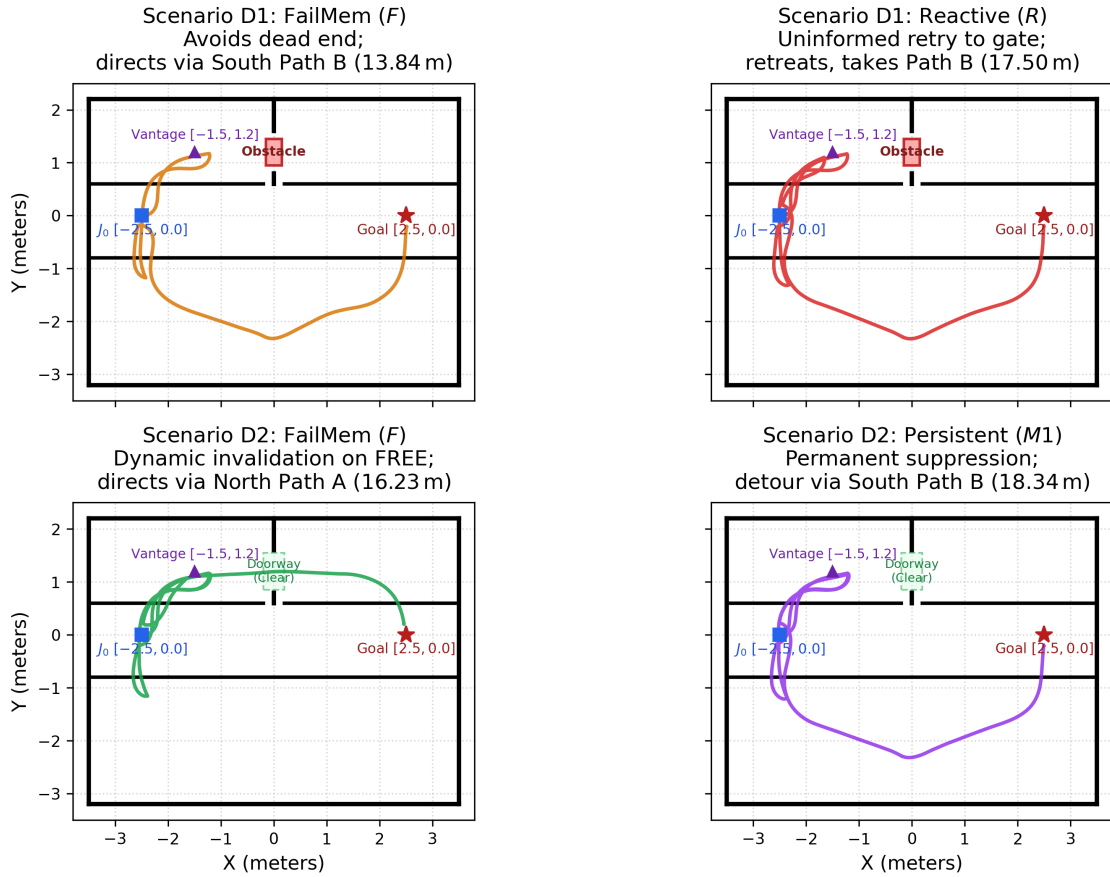
Scenario	Condition	n	Actual Route	Dead-End Traversals	Decision Dist (m)	Decision Time (s)	Total Dist (m)	Total Time (s)	Replay Audit Pass
D0	R	3	Path_A	0.0 ± 0.0	6.160 ± 0.037	51.533 ± 1.656	6.160 ± 0.037	51.533 ± 1.656	3/3 (100%)
D0	O	3	Path_A	0.0 ± 0.0	6.203 ± 0.007	49.967 ± 0.850	6.203 ± 0.007	49.967 ± 0.850	3/3 (100%)
D0	F	3	Path_A	0.0 ± 0.0	6.196 ± 0.031	49.900 ± 0.458	6.196 ± 0.031	49.900 ± 0.458	3/3 (100%)
D1	R	3	Path_A_then_Path_B	1.0 ± 0.0	10.816 ± 0.072	82.067 ± 4.394	17.220 ± 0.242	141.000 ± 5.910	3/3 (100%)
D1	O	3	Path_B	0.0 ± 0.0	7.550 ± 0.052	50.033 ± 1.986	13.947 ± 0.070	108.700 ± 2.718	3/3 (100%)
D1	F	3	Path_B	0.0 ± 0.0	7.536 ± 0.003	48.200 ± 0.954	14.027 ± 0.252	109.500 ± 1.572	3/3 (100%)
D2	R	3	Path_A	0.0 ± 0.0	5.915 ± 0.044	49.567 ± 1.595	16.344 ± 0.032	148.167 ± 3.164	3/3 (100%)
D2	O	3	Path_A	0.0 ± 0.0	5.939 ± 0.066	52.033 ± 1.747	16.396 ± 0.240	149.767 ± 0.551	3/3 (100%)
D2	F	3	Path_A	0.0 ± 0.0	5.886 ± 0.022	51.100 ± 0.954	16.247 ± 0.098	151.100 ± 7.762	3/3 (100%)
D2	$M1$	3	Path_B	0.0 ± 0.0	7.895 ± 0.023	51.300 ± 1.997	18.248 ± 0.094	148.833 ± 2.397	3/3 (100%)

5.2 Pairwise Contrasts & Trajectory Analysis

Table 2 presents key pairwise contrast comparisons across evaluated methods.

Scenario	Comparison	Metric	Method A Mean	Method B Mean	Abs Diff ($A - B$)	Rel Diff (%)
D1	F vs. R	Total Dist (m)	14.027	17.220	-3.193	-18.5%
D1	F vs. R	Total Time (s)	109.500	141.000	-31.500	-22.3%
D1	O vs. R	Total Dist (m)	13.947	17.220	-3.273	-19.0%
D1	O vs. R	Total Time (s)	108.700	141.000	-32.300	-22.9%
D1	F vs. O	Total Dist (m)	14.027	13.947	+0.080	+0.6%
D1	F vs. O	Total Time (s)	109.500	108.700	+0.800	+0.7%
D2	F vs. $M1$	Total Dist (m)	16.247	18.248	-2.000	-11.0%
D2	F vs. $M1$	Total Time (s)	151.100	148.833	+2.267	+1.5%
D2	F vs. O	Total Dist (m)	16.247	16.396	-0.149	-0.9%
D2	F vs. O	Total Time (s)	151.100	149.767	+1.333	+0.9%

Representative Physical Trajectories Across Evaluated Conditions



Representative Physical Trajectories Across Evaluated Conditions

- Elimination of Redundant Dead-End Exploration (F vs. R):** In D1, FailMem (F) eliminates redundant entrance into the blocked corridor (0.0 vs. 1.0 dead ends), reducing total distance by -18.5% and time by -22.3% relative to Reactive R . Note that R 's dead-end traversal represents an unguided retry traversal to the entrance waypoint, not physical collision or Nav2 crash.

2. **Prevention of Detour Overhead (F vs. $M1$):** In D2, dynamic invalidation upon observing FREE unsuppresses Path A, saving 2.00 m (-11.0%) total distance relative to persistent suppression ($M1$). Total execution times remain comparable (151.1 s vs. 148.8 s).
3. **Comparison with Spatial Caching in Static 2D Geometry (F vs. O):** In both D1 and D2, FailMem (F) and Spatial Cache (O) selected identical topological routes. 本次探索性数据中路线选择相同，未显示 F 的额外收益；尚未进行统计等效或一般化验证。

5.3 Feasibility Check for Action-Conditioned Navigation (H_1)

To test whether action configurations (aligned a_{aligned} vs. doorpost boundary-biased oblique a_{oblique}) create executability differences in an unobstructed doorway, a fixed 4-run feasibility check was executed (`reports/evidence/p2d_h1_feasibility/`).

Run Name	Action Profile	Target Goal	Doorway Perception	Nav2 Status (Code)	Est. Nav2 Nav Time (s)	Assumed Settling (s)	Stability Window (s)	Total Sim Duration (s)	Physical Arrival Verified
H1_aligned_run1	act_aligned	[1.50, 1.20, 0.0]	FREE (23 rays)	SUCCEEDED (4)	13.4 (est.)	3.0 (nom.)	2.4	18.8	False (excess av: 0.1068 > 0.08)
H1_aligned_run2	act_aligned	[1.50, 1.20, 0.0]	FREE (22 rays)	SUCCEEDED (4)	13.4 (est.)	3.0 (nom.)	2.4	18.8	True
H1_oblique_run1	act_oblique	[0.50, 0.88, 0.0]	FREE (23 rays)	SUCCEEDED (4)	11.1 (est.)	3.0 (nom.)	2.3	16.4	True
H1_oblique_run2	act_oblique	[0.50, 0.88, 0.0]	FREE (22 rays)	SUCCEEDED (4)	11.0 (est.)	3.0 (nom.)	2.3	16.3	True

Findings & Decision: - Nav2 returned SUCCEEDED (status code 4) in all 4/4 runs; strict physical arrival was verified in 3/4 runs (H1_aligned_run1 failed angular velocity halt threshold during the stability window). - Nav2's DWBLocalPlanner successfully navigated the doorpost inflation boundary without planner abortion in both oblique runs. - **Conclusion:** 本次候选场景未建立预期的动作可执行性差异，因此停止本轮 H_1 探索；不构成对一般动作条件失败记忆假设的证伪。

6. Threats to Validity & Limitations

1. **Exploratory Sample Size:** The 30-run comparative pilot ($n = 3$ per condition) serves as an exploratory demonstration of mechanism functioning and auditability; it does not constitute large-sample asymptotic hypothesis testing.
2. **Simulation Fidelity:** Evaluated in Gazebo 11 simulation with nominal sensor models. Hardware deployments with real-world sensor dropout, dynamic obstacles, and adversarial conditions are subject to future research.
3. **Conceptual Extensions ($F2, O+$):** Action-profiled failure memory ($F2$) and backoff spatial caching ($O+$) are formalized as conceptual models in project documentation but have not yet been evaluated in simulation.

7. Conclusion

This paper presented FailMem, an event-driven failure memory architecture for autonomous mobile robots. By binding execution failures to immutable observation bundles and providing an independent replay audit pipeline, FailMem establishes rigorous standards for reproducible robotics evaluation. Our empirical findings demonstrate that failure memory eliminates redundant dead-end exploration relative to reactive baselines and prevents permanent detour traps upon environmental recovery, while exhibiting identical route selection with spatial caching under the tested static 2D geometry. These findings define concrete experimental benchmarks and provide a solid foundation for future research in action-conditioned and multi-agent failure memory systems.

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